INCH-POUND

MIL-DTL-32434/2
23 JANUARY 2013

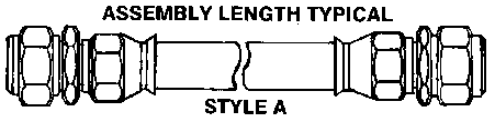
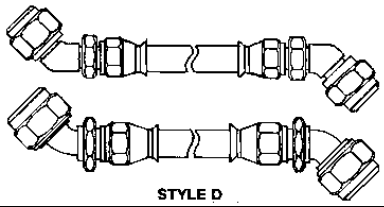
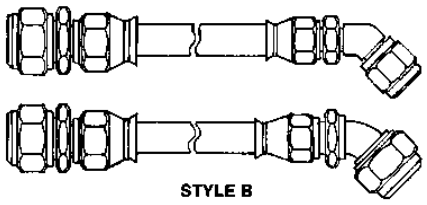
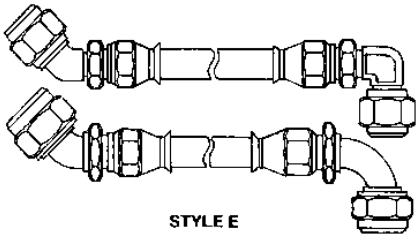
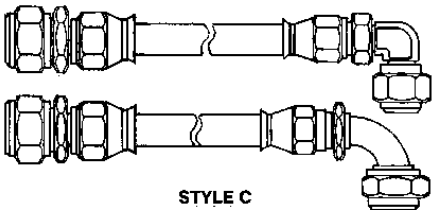
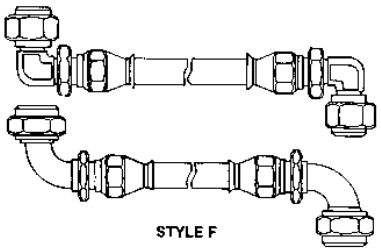
DETAIL SPECIFICATION SHEET

HOSE ASSEMBLY, POLYTETRAFLUOROETHYLENE, CONVOLUTED, REUSABLE FITTINGS, HIGH TEMPERATURE, MEDIUM PRESSURE, FLARELESS-TO-FLARELESS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32434.

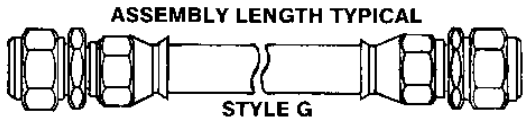
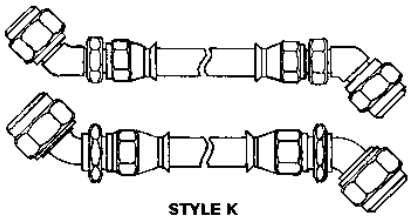
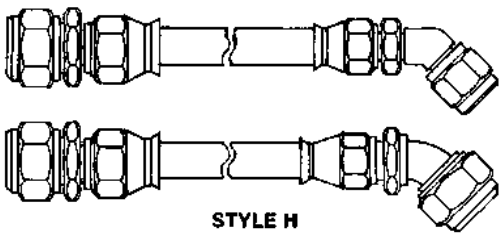
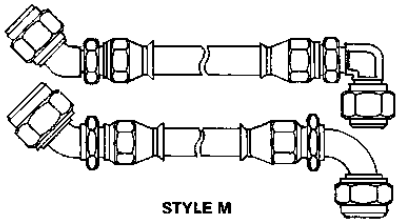
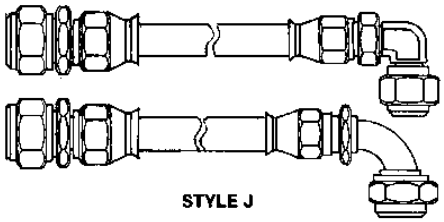
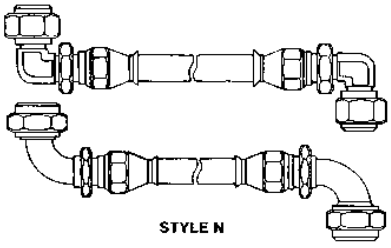
Class 1 (450°) and class 2 (275°) hose assembly styles with flareless-to-flareless reusable fittings see figures 1 and 2.

Hose assembly styles with class 1 fittings (450°F) corrosion resistant steel		Hose assembly styles with class 1 fittings (450°F) corrosion resistant steel	
End 1	End 2	End 1	End 2
 STYLE A		 STYLE D	
 STYLE B		 STYLE E	
 STYLE C		 STYLE F	

NOTES:
1. See table I for fitting ends and hose cut off factor (HCOF).
2. Fittings shall mate with parts designed to SAE-AS33514 flareless fitting end standards.

FIGURE 1. Class 1 fittings styles and designators.

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Hose assembly styles with class 2 fittings, (275°F), combination of aluminum and corrosion resistant steel		Hose assembly styles with class 2 fittings, (275°F), combination of aluminum and corrosion resistant steel	
End 1	End 2	End 1	End 2
 STYLE G		 STYLE K	
 STYLE H		 STYLE M	
 STYLE J		 STYLE N	

NOTES:
1. See tables I and II for fitting ends and HCOF.
2. Fittings shall mate with parts designed to SAE-AS33514, flareless fitting end standard.

FIGURE 2. Class 2 fittings styles and designators.

Hose cut-off factor (HCOF) for assembly with class 1 fittings: See table I.

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TABLE I. HCOF for assembly with class 1 fittings. 1/

Hose assembly styles with class 1 corrosion resistant steel (CRES) fittings 2/	Fitting ends		HCOF (size vs length) 3/									
	1	2	3/4	4	5	6	8	10	12	16	20	24
A	MS27381	MS27381	1.84	1.78	1.86	2.04	2.32	2.64	2.80	3.16	3.44	4.14
B	MS27381	MS27384	2.15	2.18	2.25	2.45	3.11					
	MS27381	MS27382						3.09	3.63	3.91	4.33	5.10
C	MS27381	MS27385	1.78	1.80	1.90	2.05	2.47					
	MS27381	MS27383						2.73	3.32	3.63	4.06	4.75
D	MS27384	MS27384	2.46	2.58	2.64	2.86	3.90					
	MS27382	MS27382						3.54	4.46	4.66	5.22	6.06
E	MS27384	MS27385	2.09	2.20	2.29	2.46	3.26					
	MS27382	MS27383						3.18	4.15	4.38	4.95	5.71
F	MS27385	MS27385	1.72	1.82	1.94	2.06	2.62					
	MS27383	MS27383						2.82	3.84	4.10	4.68	5.36

1/ Class 1 fittings, styles A through F, are made from CRES.
2/ For depiction of styles see figure 1.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce a 20.50 inch length assembly of size 6, style A is calculated as follows: 20.50 - 2.04 = 18.46.

HCOF for assembly with class 2 fittings: See table II.

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TABLE II. HCOF for assembly with class 2 fittings. 1/

Hose Assembly Styles with class 2 combination of aluminum and CRES fittings 2/	Fitting ends		HCOF (size vs length) 3/					
	1	2	8	10	12	16	20	24
G	MS27381	MS27381	2.32	2.64	2.80	3.16	3.44	4.14
H	MS27381	MS27384	3.11					
	MS27381	MS27382		3.09	3.63	3.91	4.33	5.10
J	MS27381	MS27385	2.47					
	MS27381	MS27383		2.73	3.32	3.63	4.06	4.75
K	MS27384	MS27384	3.90					
	MS27382	MS27382		3.54	4.46	4.66	5.22	6.06
M	MS27384	MS27385	3.26					
	MS27382	MS27383		3.18	4.15	4.38	4.95	5.71
N	MS27385	MS27385	2.62					
	MS27383	MS27383		2.82	3.84	4.10	4.68	5.36

1/ Class 2 fittings, styles G through N, are made from combination of aluminum and CRES.
2/ For depiction of styles see figure 1.
3/ The HCOF is used in the following calculation to determine the hose length required to produce an assembly of a specific size, style and length: Assembly length - HCOF = Hose length. For example, the hose length required to produce a 20.50 inch length assembly of size 10, style J is calculated as follows: 20.50 - 2.73 = 17.77.

REQUIREMENTS

Assembly classification: Class 1 and class 2 hose assemblies, as specified in MIL-DTL-32434, have been incorporated into the Part or Identifying Number (PIN) as a part of styles (see tables I and II and figure 1).

Dimensions: Unless otherwise specified, all dimensions are in inches.

Fittings: All fittings shall be qualified in accordance with MIL-DTL-27272.

Flareless fitting, hose connector design: If NAS1760 flareless fitting end design is used in place of SAE-AS33514 in hose assembly design, see MIL-DTL-25579/1 for application and construction.

Hose: The hose shall be in accordance with MIL-DTL-32435.

Hose size code and elbow drop height: See table III.

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TABLE III. Size code and elbow fitting drop height.

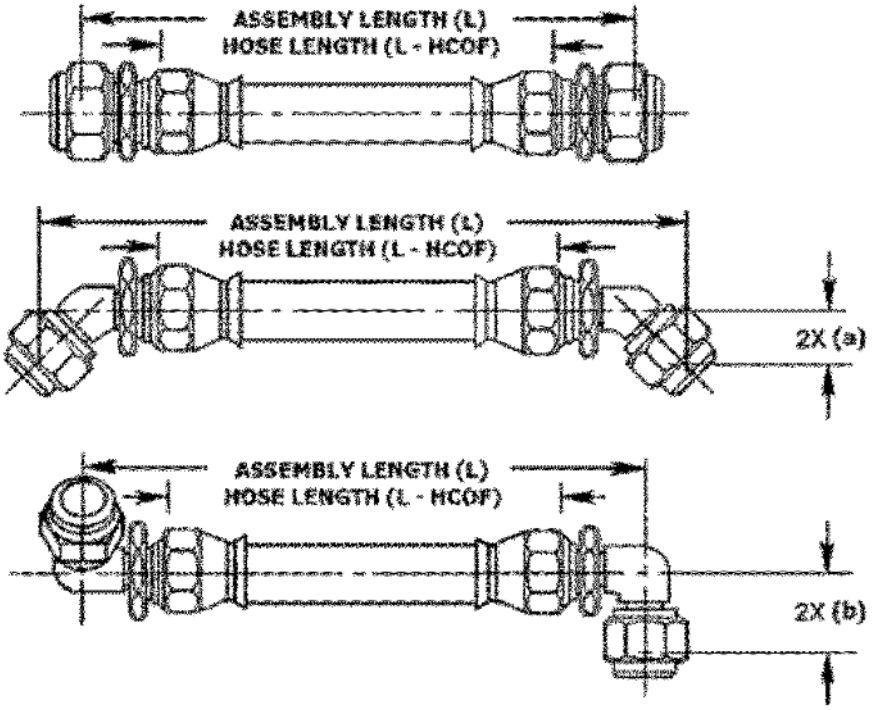
Size	Reference tube OD	Size code	Maximum drop height of elbow fitting	
			(a) 1/	(b) 1/
3/4	.250 2/	B	.452	.761
4	.250	E	.444	.745
5	.313	F	.468	.828
6	.375	G	.546	.935
8	.500	H	.642	1.072
10	.625	J	.760	1.427
12	.750	K	.835	1.661
16Z	1.000	M	.889	1.811
20Z	1.250	N	.997	2.091
24Z	1.500	P	1.190	2.473

1/ Dimensions (a) and (b) are shown on figure 3.
2/ Size 3/4 hose assembly consists of a size 4 tube OD hose assembled with size 3 tube OD fittings.

Assembly length and tolerance: Hose assembly shall be furnished in lengths as specified in the contract or purchase order (see figure 3); however, tolerances on the length of each hose assembly shall be as follows:

a. ±1/8 inch for lengths under 18 inches.
b. ±1/4 inch for lengths from 18 inches to 36 inches.
c. ±1/2 inch for lengths from 36 inches to 50 inches.
d. ±1% for lengths over 50 inches.

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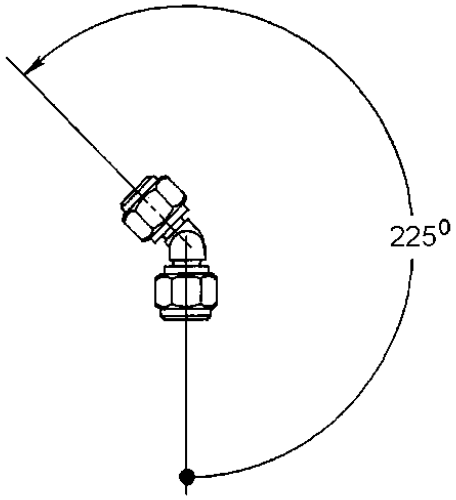
NOTES:

1. Hose assembly length “L” shall be measured, with the hose laid out horizontally and straight between the centers of the MS8000 series nipple ends, along a straight line parallel to the hose length (see MIL-DTL-25579/1).
2. See table III for dimensions of (a) and (b). Dimensions (a) and (b) are measured to the end of a MS8000 series style fitting, see MIL-DTL-25579/1. Adjustments should be made for a NAS1760 style fitting.

FIGURE 3. Elbow fitting drop height, dimensions (a) and (b).

Angular alignment: Hose assemblies with elbow fittings on each end shall have the angular orientation between the elbows measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o'clock, to the centerline of the other fitting (see figure 4). The elbow fitting drop height shall be as shown on figure 3 and table III. When applicable, the angular alignment shall be expressed in three digits and specified in the PIN.

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NOTE: Angular alignment shall be measured in degrees with a tolerance of ±2°.

FIGURE 4. Measurement of angular alignment between elbow fittings.

Protective sleeve: If required, the hose assembly shall include a protective sleeve (see table IV) and its code shall be included in the PIN.

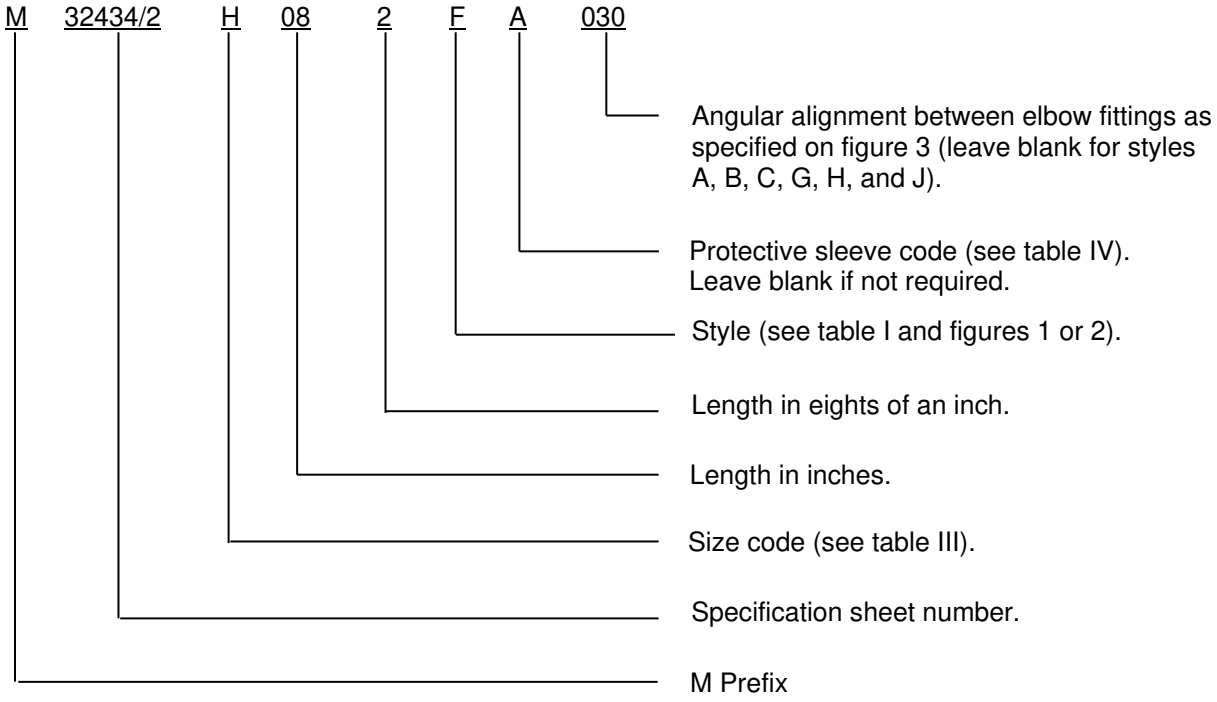
TABLE IV. Protective sleeve code.

Code	Type
Blank	None
A	SAE-AS1072 sleeve, fire protection, silicone covered, temperature ranging from -65°F to 500°F, secured with CRES bands as required. 1/
B	SAE-AS1073 - code B sleeve, abrasion protection, heat shrinkable, black polyolefin, temperature ranging from -65°F to 250 °F.
C	SAE-AS1291 - code A sleeve, chafe guard, extruded seamless white PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required.
D	SAE-AS1291 - code C sleeve, chafe guard, extruded seamless transparent FEP, temperature ranging from -65°F to 350°F, secured with CRES bands as required.
E	SAE-AS1298 sleeve, heavy wall chafe guard, extruded seamless black PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required.
L	Lock-wire hole
F	Code A + L
G	Code B + L
H	Code C + L
J	Code D + L
K	Code E + L

1/ To prevent wicking of fluids, the cut end of the fire protective sleeve (code A) shall be coated with room temperature vulcanized (RTV) silicone rubber prior to installation. After installation, cracks and voids in the fire protective sleeve shall be coated with RTV rubber to prevent exposure of fiberglass.

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PIN: The PIN consists of letter M, specification sheet number, hose size code, hose length in inches and an eighths of an inch, hose assembly style, sleeve code, and angular alignment in degree between elbow fittings.



PIN example: M32434/2H082FA030 represents ½ inch OD PTFE convoluted hose assembly, 8.25 inches long, flare-to-flange reusable fitting with fire protective sleeve and 30° angular alignment between elbows.

Referenced documents. In addition to MIL-DTL-32434, this document references the following:

MIL-DTL-27272	MS27384
MIL-DTL-25579/1	MS8000
MIL-DTL-32435	NAS1760
MS27385	SAE-AS1072
MS27381	SAE-AS1073
MS27382	SAE-AS1291
MS27383	SAE-AS1298
	SAE-AS33514

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CONCLUDING MATERIAL

Custodians:
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Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2012-020)

Review activities:
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Navy - MC, SA, SH
Air Force - 70

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